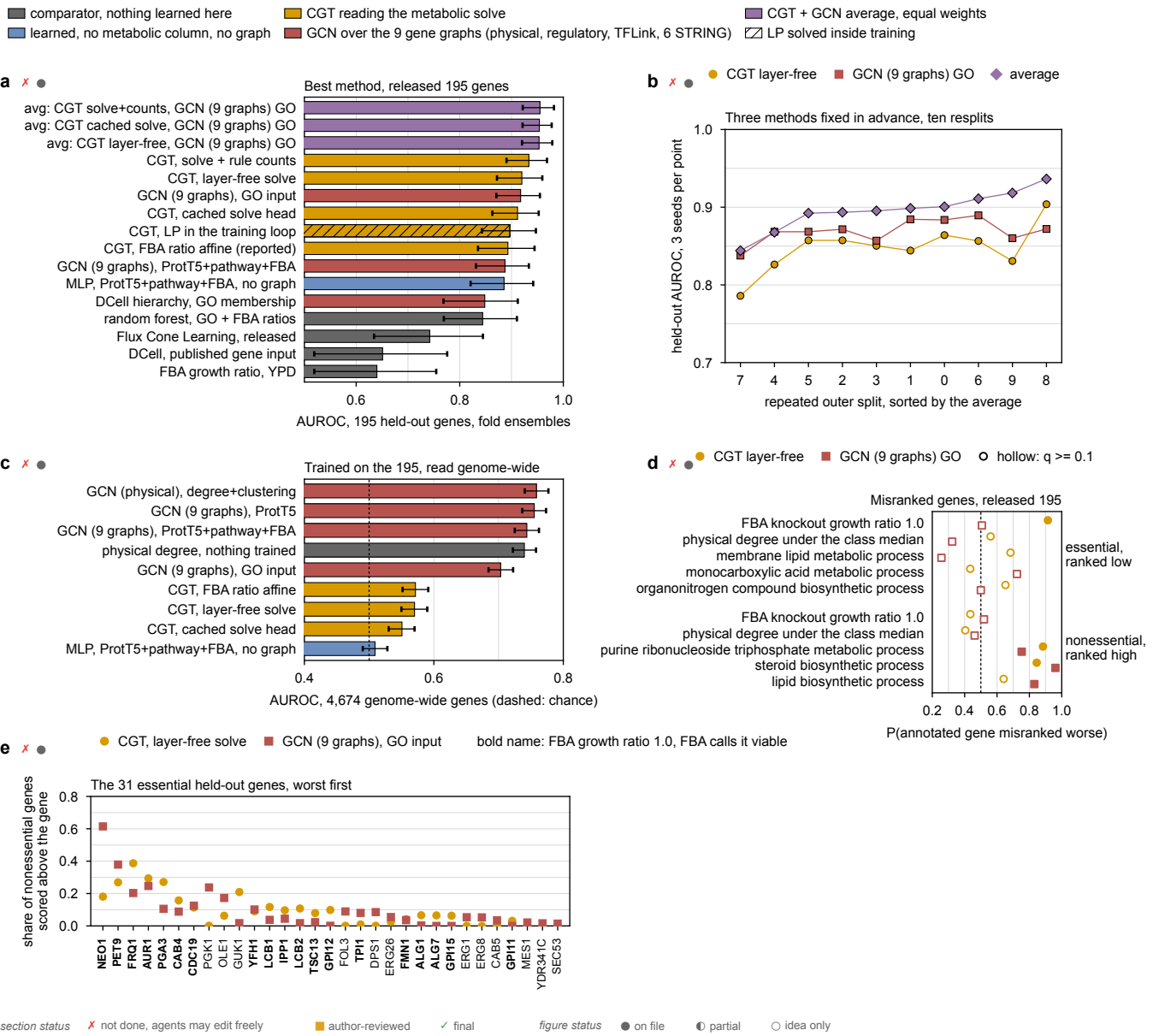

comparator, nothing learned here
learned, no metabolic column, no graph
CGT reading the metabolic solve
GCN over the 9 gene graphs (physical, regulatory, TFLink, 6 STRING)
CGT + GCN average, equal weights
LP solved inside training
a
Best method, released 195 genes
avg: CGT solve+counts, GCN (9 graphs) GO
avg: CGT cached solve, GCN (9 graphs) GO
avg: CGT layer-free, GCN (9 graphs) GO
CGT, solve + rule counts
CGT, layer-free solve
GCN (9 graphs), GO input
CGT, cached solve head
CGT, LP in the training loop
CGT, FBA ratio affine (reported)
GCN (9 graphs), ProtT5+pathway+FBA
MLP, ProtT5+pathway+FBA, no graph
DCell hierarchy, GO membership
random forest, GO + FBA ratios
Flux Cone Learning, released
DCell, published gene input
FBA growth ratio, YPD
0.6
0.8
1.0
AUROC, 195 held-out genes, fold ensembles
b
CGT layer-free
GCN (9 graphs) GO
average
Three methods fixed in advance, ten resplits
held-out AUROC, 3 seeds per point
1.0
0.9
0.8
0.7
7 4 5 2 3 1 0 6 9 8
repeated outer split, sorted by the average
c
Trained on the 195, read genome-wide
GCN (physical), degree+clustering
GCN (9 graphs), ProtT5
GCN (9 graphs), ProtT5+pathway+FBA
physical degree, nothing trained
GCN (9 graphs), GO input
CGT, FBA ratio affine
CGT, layer-free solve
CGT, cached solve head
MLP, ProtT5+pathway+FBA, no graph
0.4
0.6
0.8
AUROC, 4,674 genome-wide genes (dashed: chance)
d
CGT layer-free
GCN (9 graphs) GO
hollow: q >= 0.1
Misranked genes, released 195
FBA knockout growth ratio 1.0
physical degree under the class median
membrane lipid metabolic process
monocarboxylic acid metabolic process
organonitrogen compound biosynthetic process
FBA knockout growth ratio 1.0
physical degree under the class median
purine ribonucleoside triphosphate metabolic process
steroid biosynthetic process
lipid biosynthetic process
essential, ranked low
nonessential, ranked high
0.2
0.4
0.6
0.8
1.0
P(annotated gene misranked worse)
e
CGT, layer-free solve
GCN (9 graphs), GO input
bold name: FBA growth ratio 1.0, FBA calls it viable
The 31 essential held-out genes, worst first
share of nonessential genes scored above the gene
0.8
0.6
0.4
0.2
0.0
NEO1 PET9 FRQ1 AUR1 PGA3 CAB4 CDC19 PGK1 OLE1 GUK1 YFH1 LCB1 IPP1 LCB2 TSC13 GPI12 FOL3 TPI1 DPS1 ERG26 FMN1 ALG1 ALG7 GPI15 ERG1 ERG8 CAB5 GPI11 MES1 YDR341C SEC53
section status
✗ not done, agents may edit freely
■ author-reviewed
✓ final
figure status
● on file
◐ partial
○ idea only